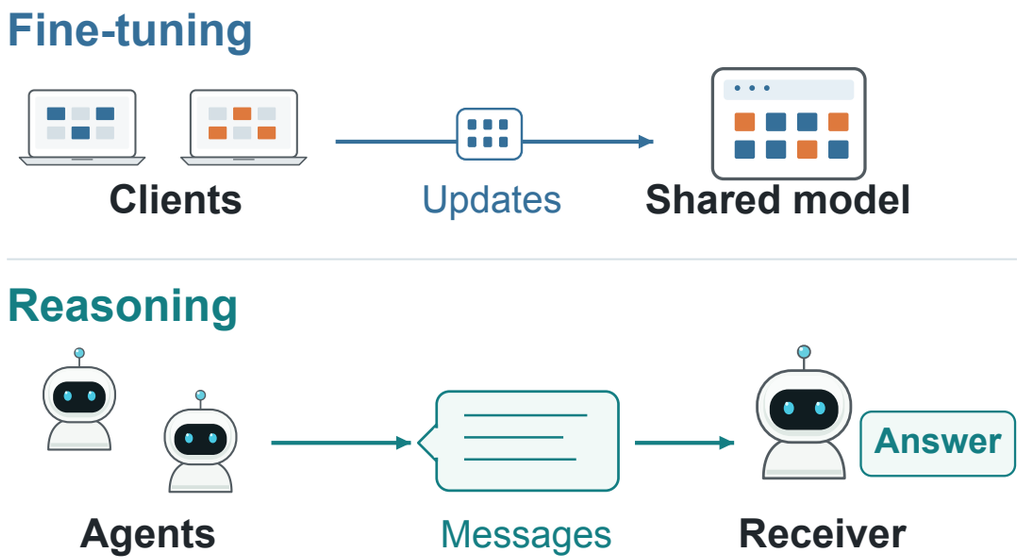
SFU SIMON FRASER UNIVERSITY

Rethinking Collaboration: Cost and Value in Fine-Tuning and Multi-Agent Reasoning

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November 2026

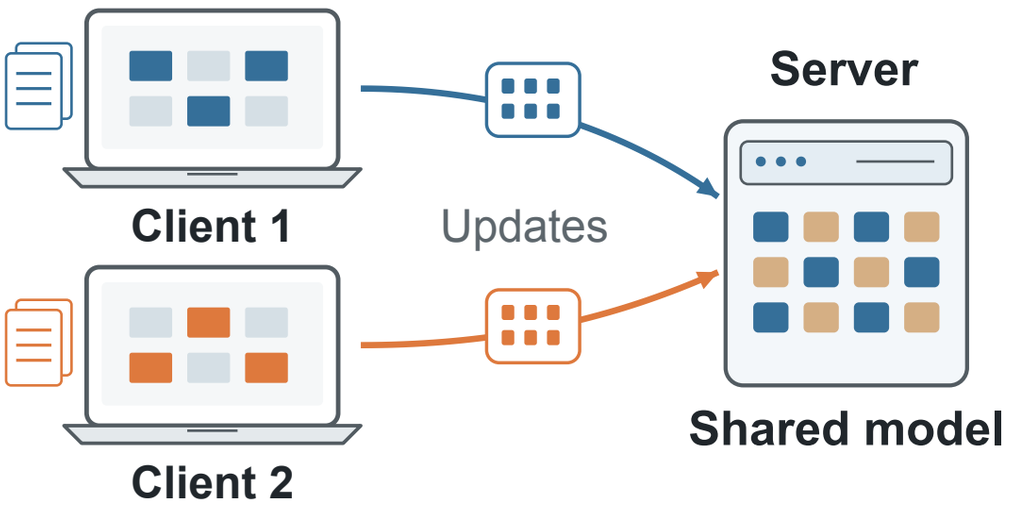


Fine-tuning and reasoning

During fine-tuning

Learn from clients' local data.

Model weights are updated



1 What to update 2 How to combine

Value: learning performance

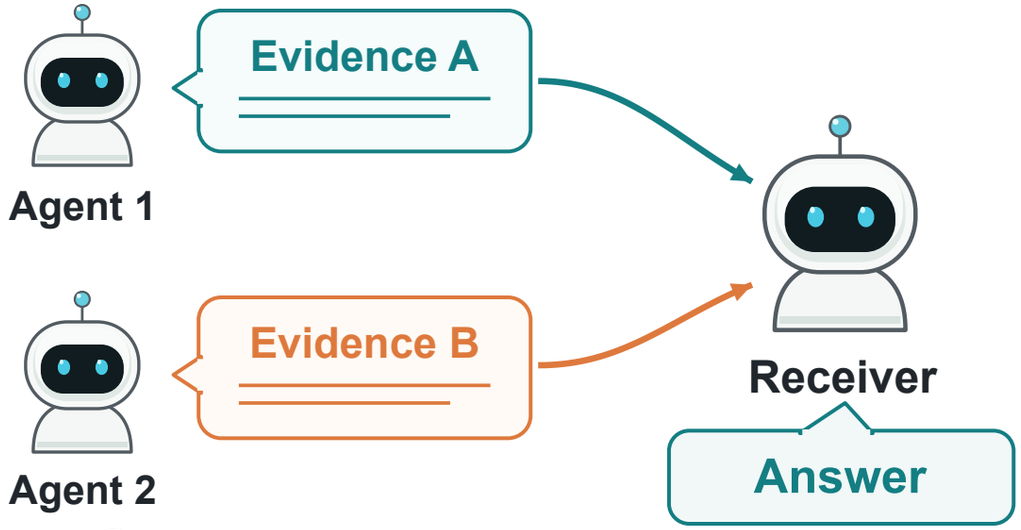
Cost: memory, training, communication

ITU-T Y.3193 (2026) · IMT-2030

During reasoning

Use evidence held by other agents.

Model weights stay fixed



3 Which messages to use

Value: message & answer quality

Cost: generation, receiver runs, scoring

A2A: communication across agent systems

When information can be exchanged, how do we make that exchange worthwhile?

Three stages & Questions

1

At the client

Fine-tuning

How can each client update its model with limited memory?

2

At the server

Fine-tuning

How should the server combine the updates it receives?

3

At the receiving agent

Inference

Which messages should the agent use to form its answer?

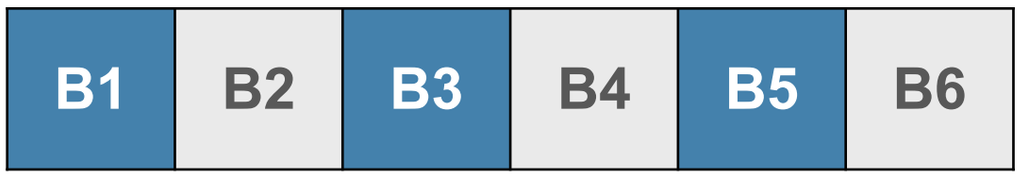
Fine-tuning changes model weights; inference changes the answer.

What can a client train with the limited GPU memory?

Fine-tuning · Client

When clients cannot fully train a model:

A client model



Blue: updated Gray: frozen

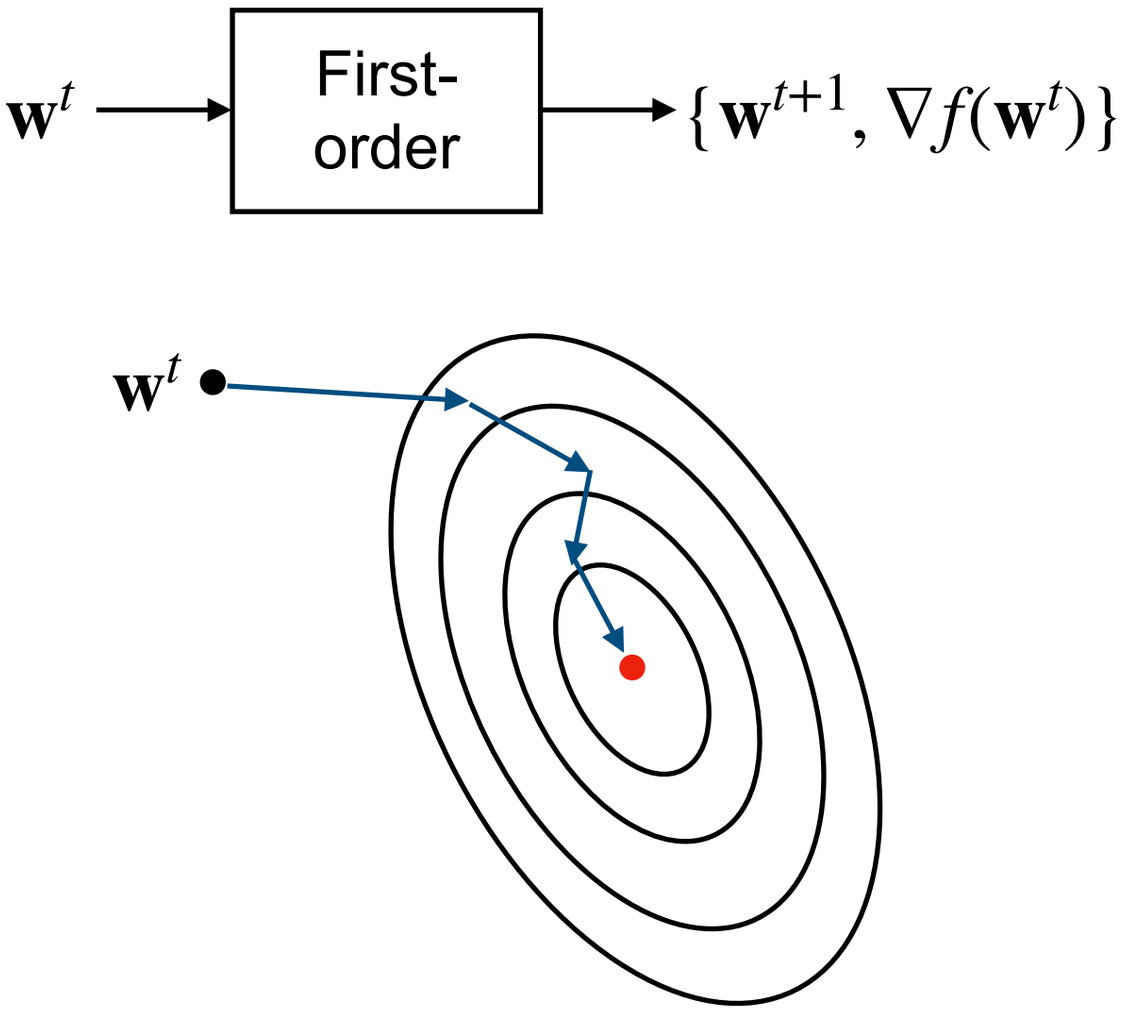
Fewer updated blocks

Lower activation memory, but block allocation also affects convergence speed and learning performance.

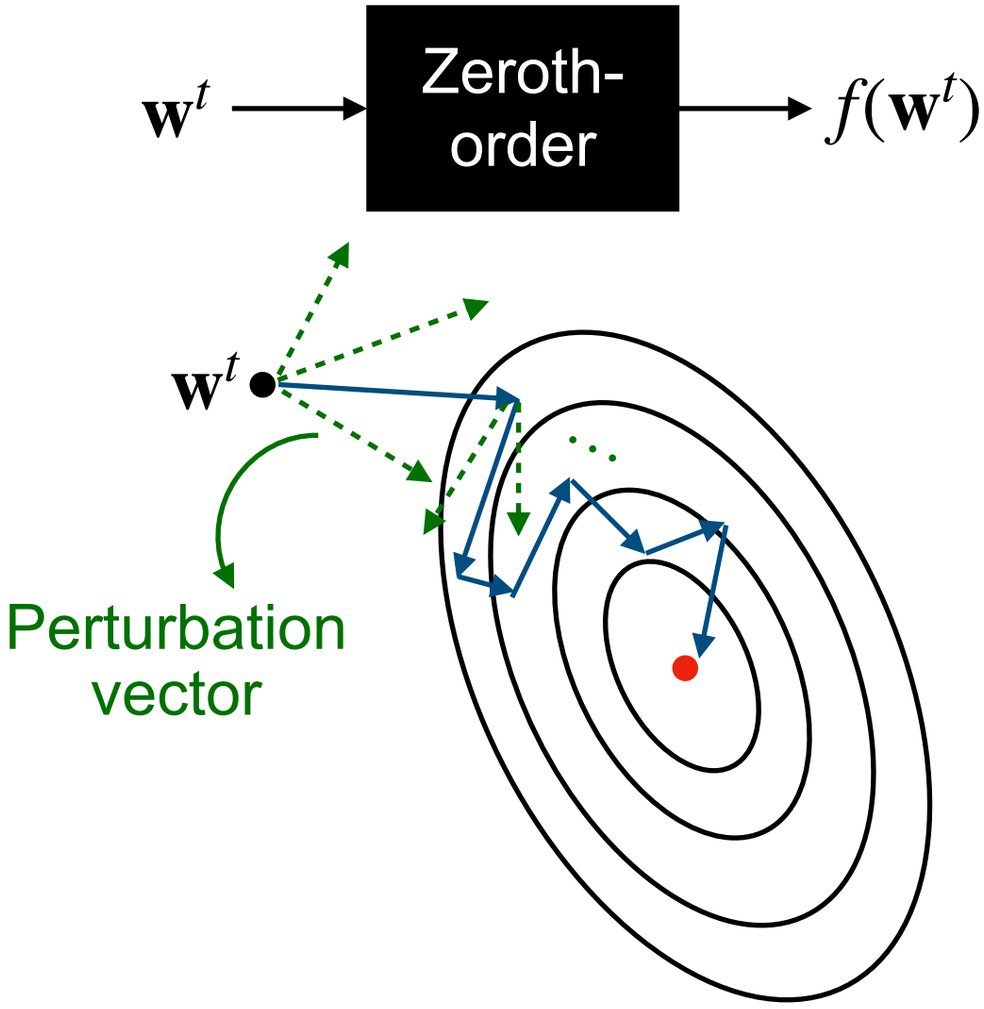
Besides partial model update, is there any other approach to further save memory?

Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. “ZorBA: Zeroth-order Federated Fine-tuning of LLMs with Heterogeneous Block Activation.” In *Proceedings of the IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan, May 2026.

First-order and zeroth-order optimization



Gradients are computed through backpropagation.



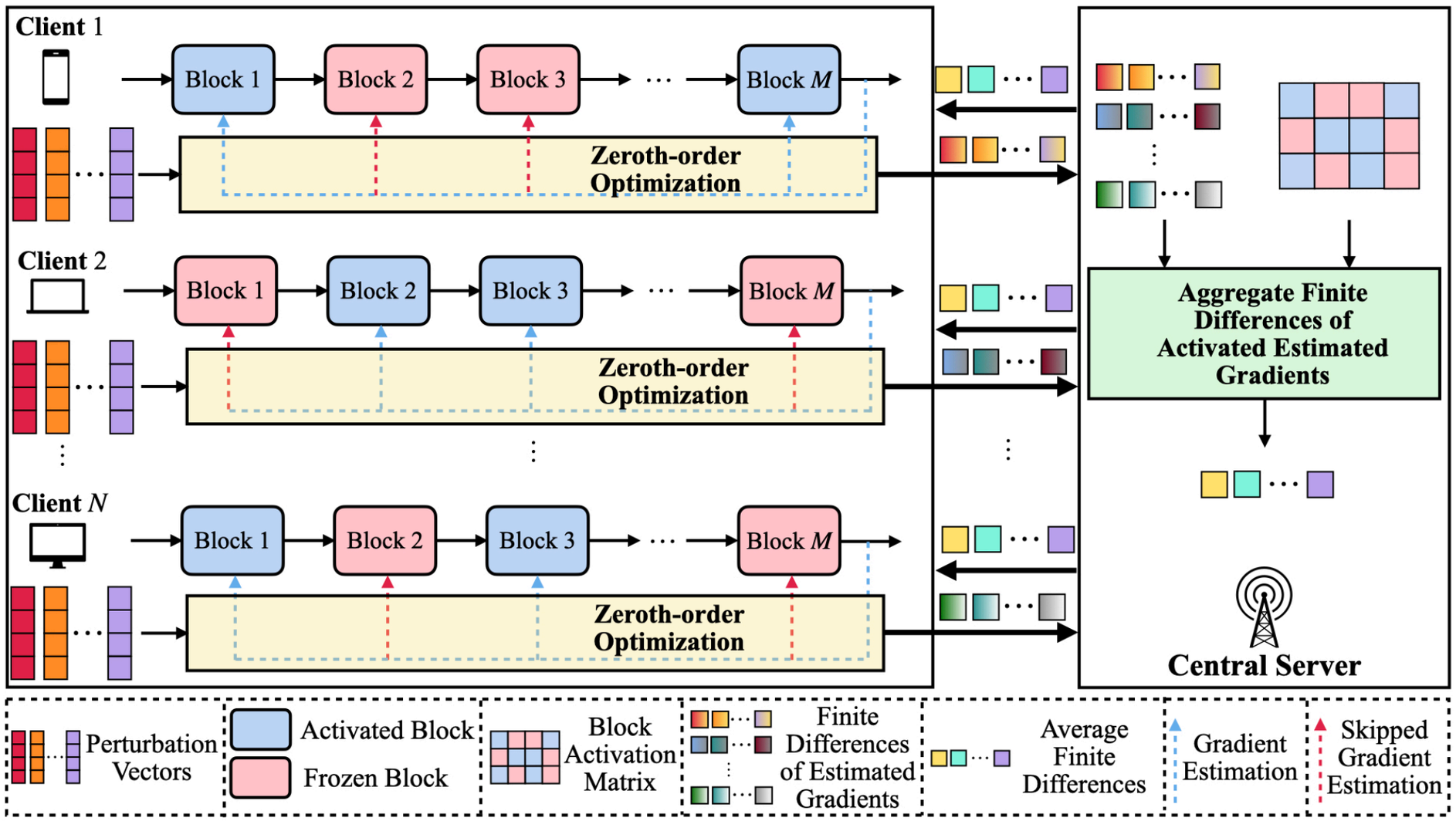
Gradients are estimated from perturbed forward evaluations.

Further save memory by eliminating the need to store gradients.

Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. "ZorBA: Zeroth-order Federated Fine-tuning of LLMs with Heterogeneous Block Activation." In *Proceedings of the IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan, May 2026.

Allocate blocks to fit heterogeneous clients

ZorBA: selective activation and zeroth-order optimization



Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. "ZorBA: Zeroth-order Federated Fine-tuning of LLMs with Heterogeneous Block Activation." In *Proceedings of the IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan, May 2026.

Less memory can require more rounds

OPT-1.3B · AG-News · 80% target accuracy

Method	Total client VRAM
FedIT	1,431.17 GB
FedZO	1,189.03 GB
DeComFL	1,189.03 GB
ZorBA	538.04 GB

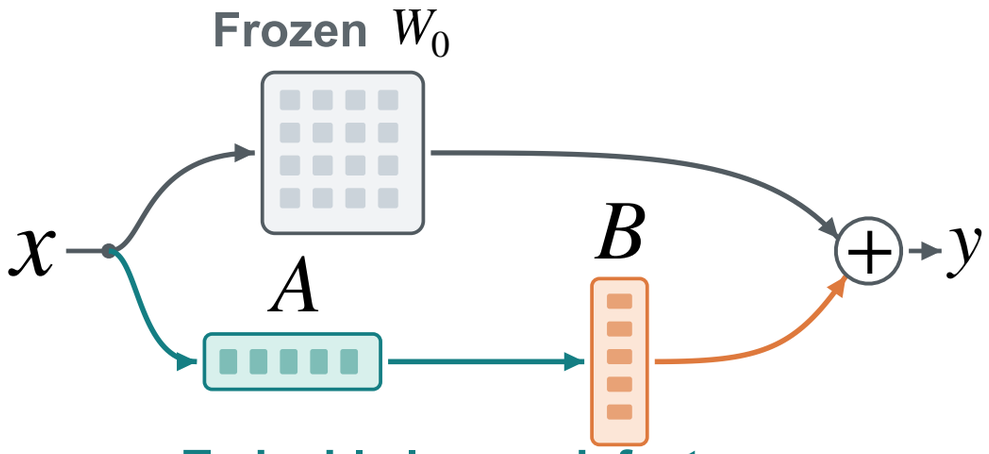
Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. “ZorBA: Zeroth-order Federated Fine-tuning of LLMs with Heterogeneous Block Activation.” In *Proceedings of the IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan, May 2026.

Federated LoRA: what should the server combine?

Fine-tuning · Server

Inside each client: LoRA

Low-rank adaptation

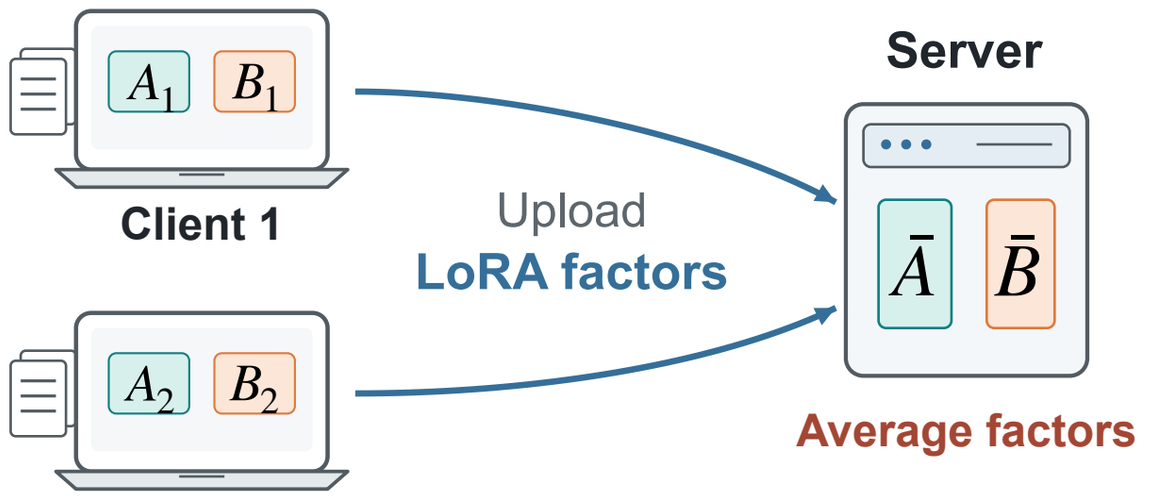


Trainable low-rank factors

LoRA update: $\Delta W = BA$

Across clients: share the factors

Local data stays with each client.



Scalar example · equal client weights

Client	B	A	BA
1	1	1	+1
2	−1	−1	+1

Average updates

$$\frac{1+1}{2} = 1$$

Local update preserved

Average factors

$$0 \times 0 = 0$$

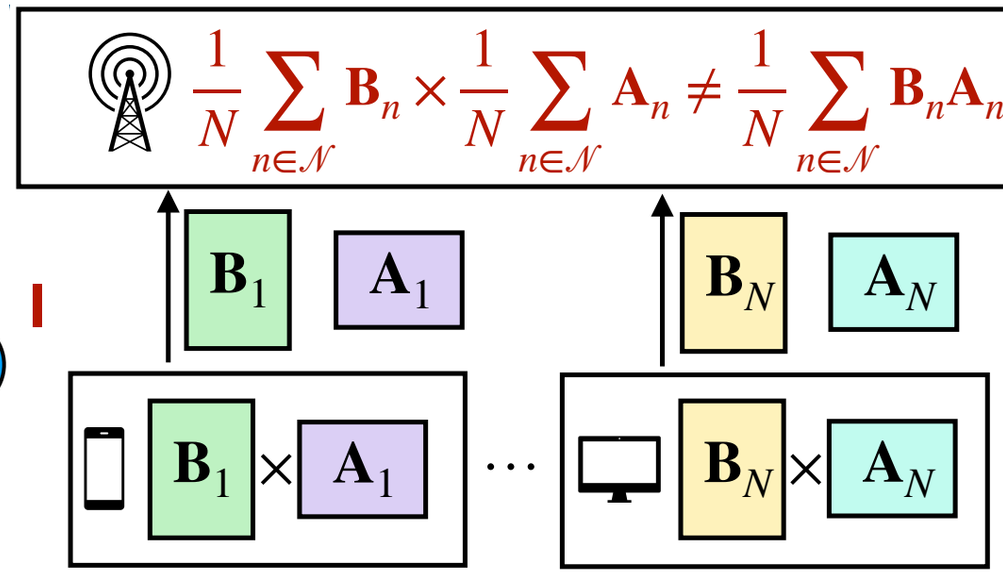
Both factor averages are zero

The server can lose an update that both clients learned.

Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. “FLoRG: Federated Fine-tuning with Low-rank Gram Matrices and Procrustes Alignment.” In *Proceedings of the International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, Apr. 2026.

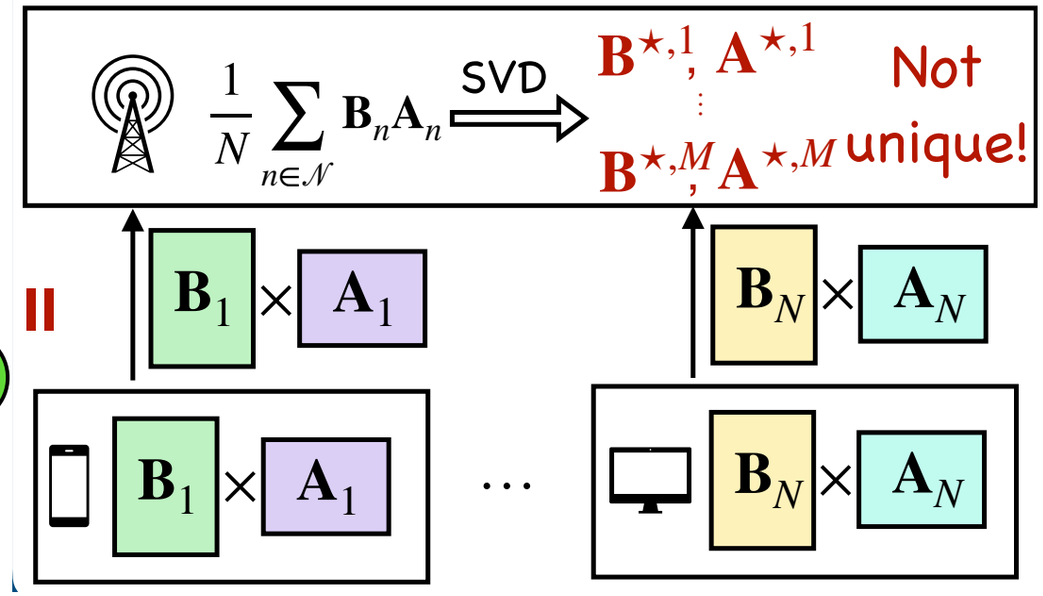
What are the challenges within?

Aggregation Error



Separate factor averages can change the update.

Non-unique Decomposition

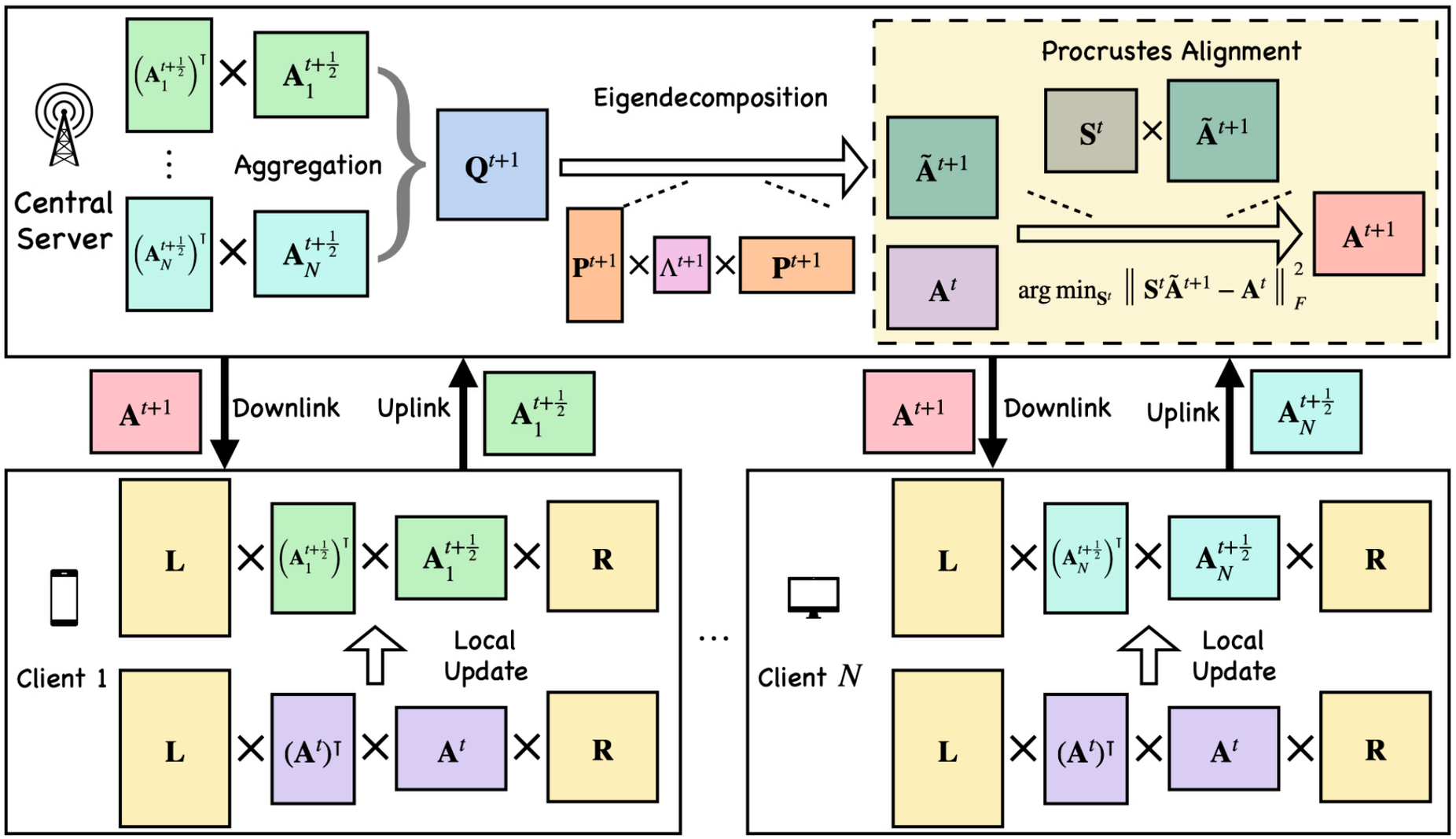


A correct aggregation still has multiple factor representations.

Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. “FLoRG: Federated Fine-tuning with Low-rank Gram Matrices and Procrustes Alignment.” In *Proceedings of the International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, Apr. 2026.

Aggregate in the Gram representation

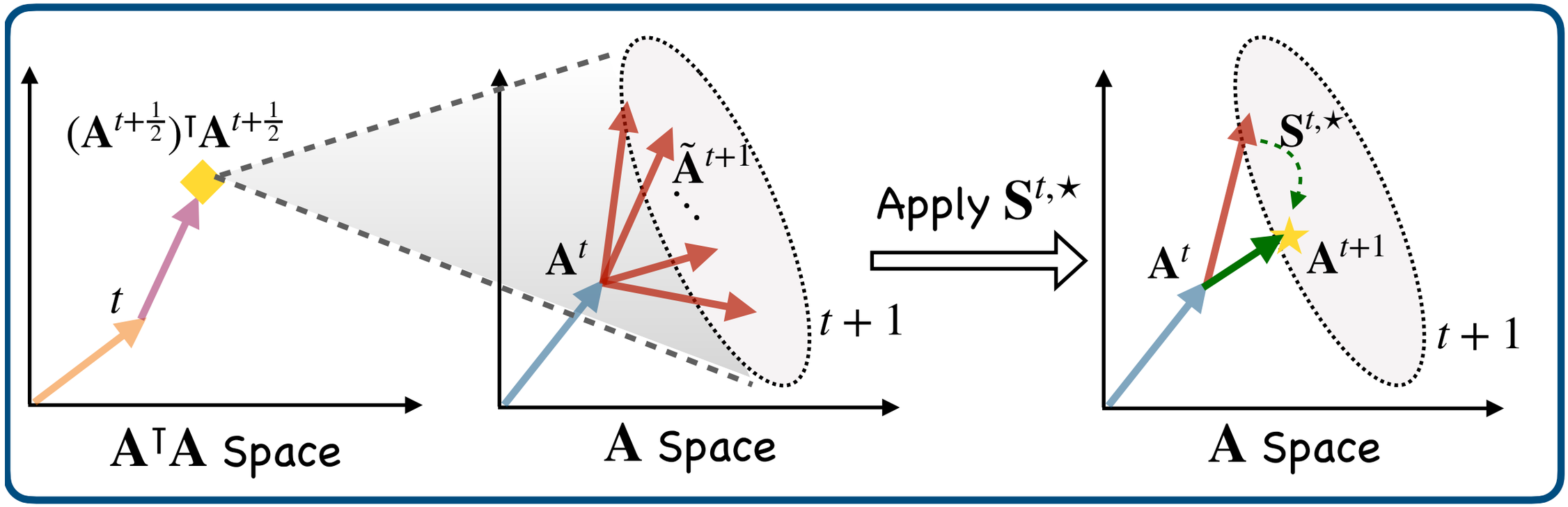
FLoRG: $\Delta W = LA^{\top}AR$



Transmit compact factors; combine their Gram matrices.

Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. “FLoRG: Federated Fine-tuning with Low-rank Gram Matrices and Procrustes Alignment.” In *Proceedings of the International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, Apr. 2026.

Procrustes alignment in parameter subspace



Equivalent factorizations

1 The dashed set contains factors that share the same Gram matrix.

Orientation closest to A^t

2 Orthogonal alignment preserves the Gram matrix and limits factor drift.

Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. "FLoRG: Federated Fine-tuning with Low-rank Gram Matrices and Procrustes Alignment." In *Proceedings of the International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, Apr. 2026.

Communication should be measured to a target

RoBERTa-large · QNLI · 85% target accuracy

Method	Transmitted parameters
FedIT	81.2 M
FeDeRA	51.7 M
FFA-LoRA	33.5 M
FedSA-LoRA	44.2 M
FedEx-LoRA	29,600 M
FLoRG	14.5 M

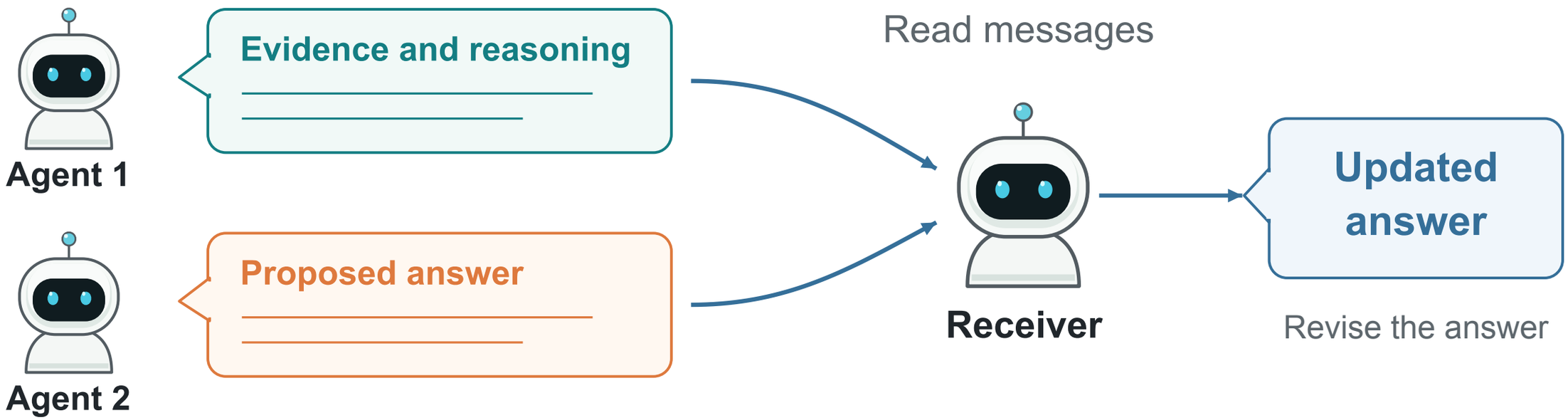
Compact aggregation reduces the cost of reaching the same target.

Chuiyang Meng, Ming Tang, and Vincent W.S. Wong. “FLoRG: Federated Fine-tuning with Low-rank Gram Matrices and Procrustes Alignment.” In *Proceedings of the International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, Apr. 2026.

Multi-agent reasoning: which messages help?

Inference · Receiver

Same task, fixed model weights



1 **Measure the message effect**

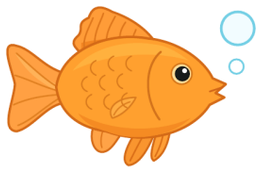
Does it improve this receiver's answer?

2 **Select useful messages**

Which messages are worth their inference cost?

Chuiyang Meng, Wenlu Yu, Ming Tang, and Cheng Li. "Social Circuits Behind Multi-Agent Echo Chambers." Manuscript, 2026. Submitted to ICLR 2027.

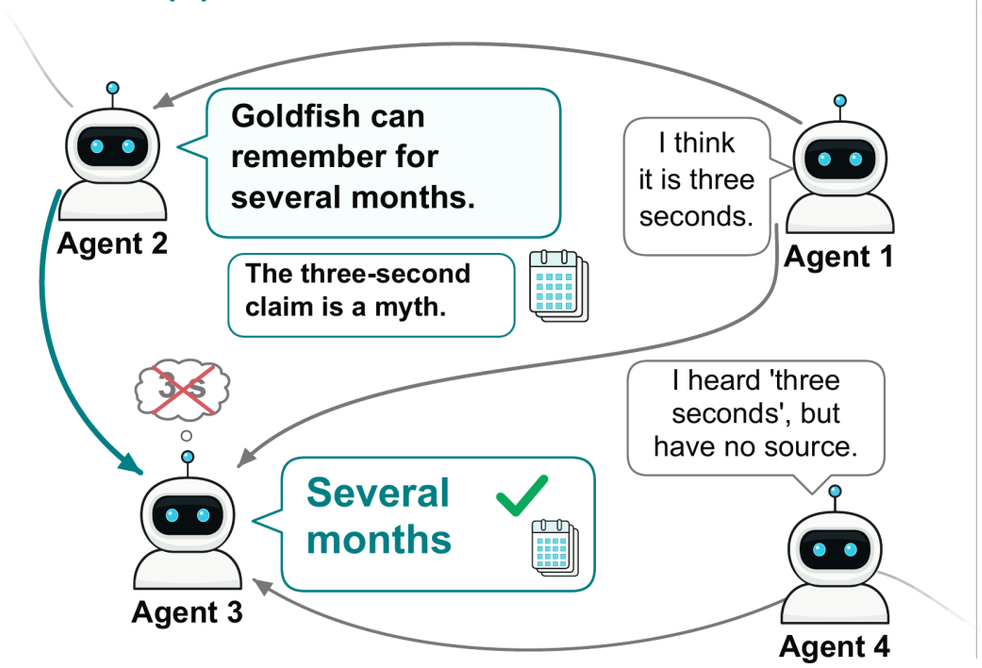
Messages can correct or reinforce a shared error

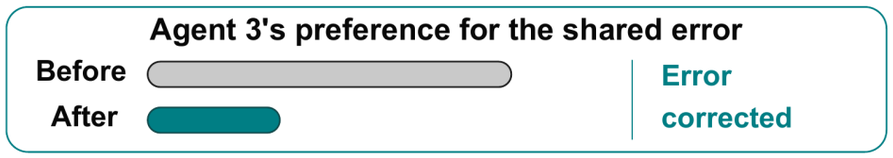


How long is a goldfish's memory?

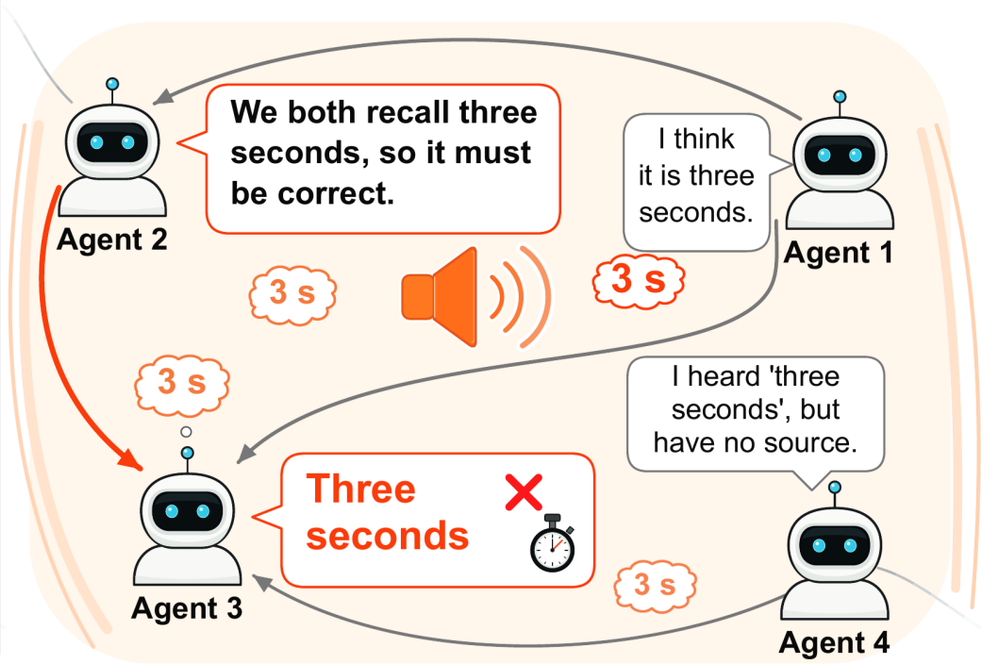


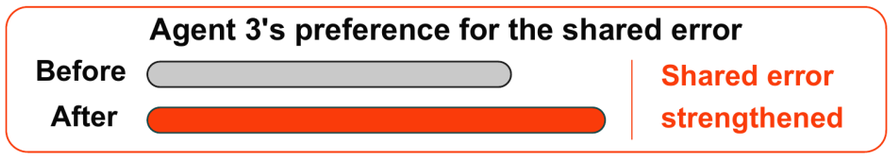
(a) Correct an error





(b) Reinforce an echo chamber



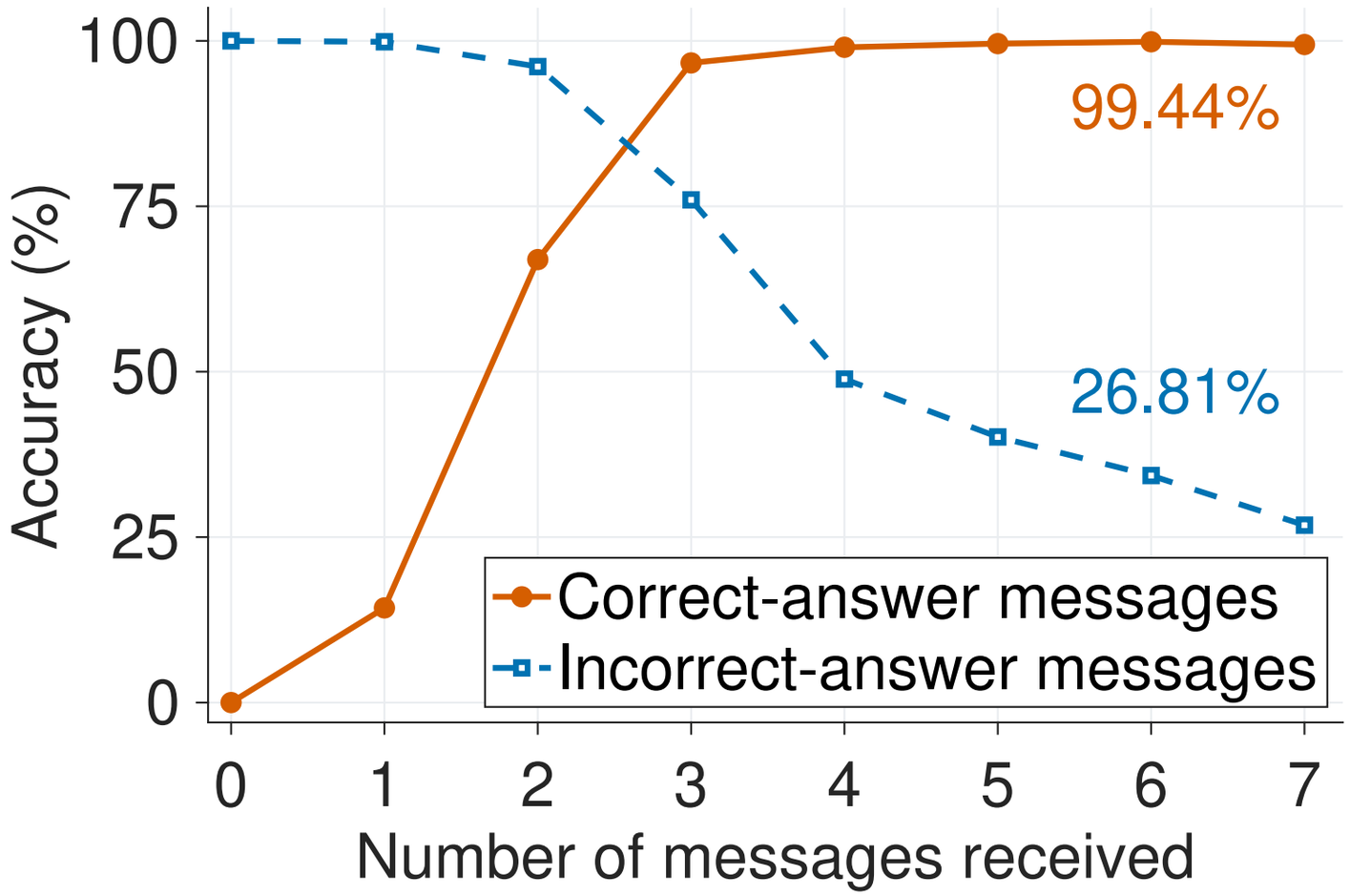


Why it happens? What does the message change inside the receiver?

Chuiyang Meng, Wenlu Yu, Ming Tang, and Cheng Li. “Social Circuits Behind Multi-Agent Echo Chambers.” Manuscript, 2026. Submitted to ICLR 2027.

More messages can improve or degrade accuracy

Gemma-3-4B-it · Two message conditions



Observation

Sending more messages can strengthen either a correct answer or an error.

What's Needed

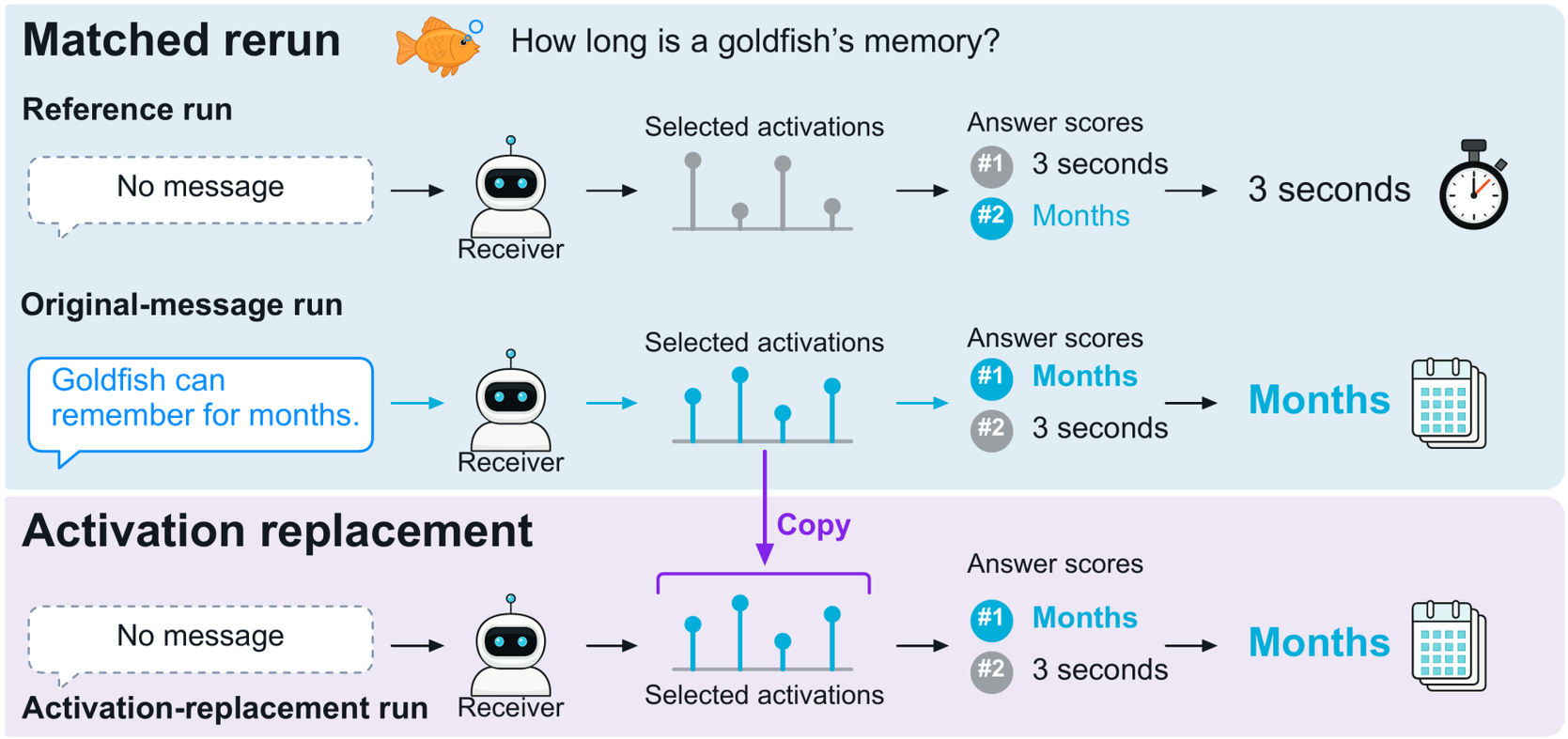
Identify a message's effect on the receiving agent.

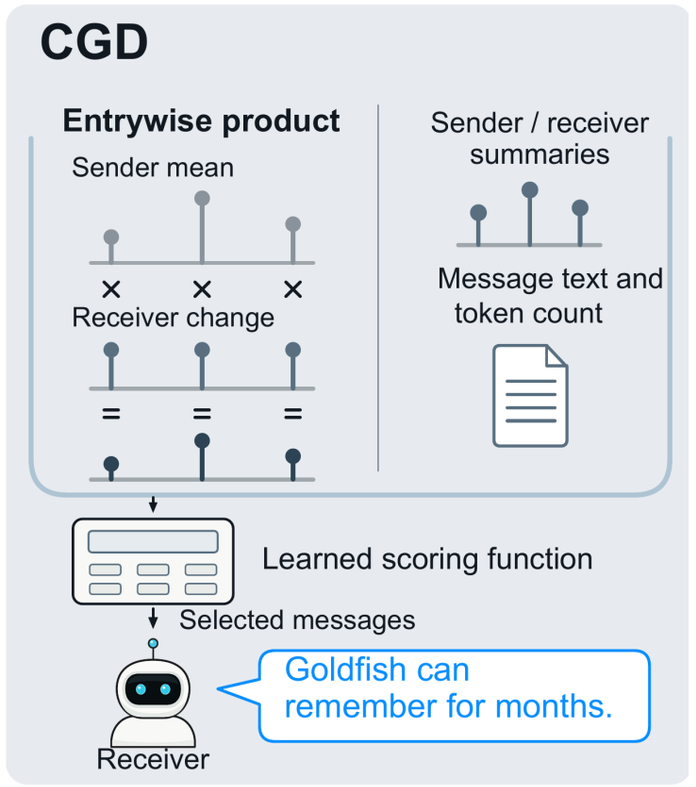
Agreement does not determine the collaboration value. Quality matters.

Chuiyang Meng, Wenlu Yu, Ming Tang, and Cheng Li. "Social Circuits Behind Multi-Agent Echo Chambers." Manuscript, 2026. Submitted to ICLR 2027.

Trace the message effect through the receiver

We use message reruns and activation replacement isolate message effects.





Message → **Activation** → **Answer**

Chuiyang Meng, Wenlu Yu, Ming Tang, and Cheng Li. "Social Circuits Behind Multi-Agent Echo Chambers." Manuscript, 2026. Submitted to ICLR 2027.

CGD improves accuracy under matched token limits

2Wiki free generation · 240 tasks · matched total token limits

Model	Strongest baseline	CGD accuracy
Gemma-3-4B	Majority vote 26.39 ± 2.77	42.92 ± 0.83
Qwen3-4B	SafeSieve 40.56 ± 2.37	51.81 ± 0.64
Qwen3-8B	MAD-M2 30.83 ± 0.83	61.53 ± 0.48

Chuiyang Meng, Wenlu Yu, Ming Tang, and Cheng Li. “Social Circuits Behind Multi-Agent Echo Chambers.” Manuscript, 2026. Submitted to ICLR 2027.

Takeaways

During training

1 Distributed clients can collaborate. What the shared model learns depends on how that work is divided and how updates are combined.

During reasoning

2 A message can correct a mistake or reinforce it.
Its value depends on how it changes the internal activations.

When should we learn, and when should we deliberate?

When the same mistake keeps recurring

1. Can the model learn from past corrections and adapt to future tasks on its own?

When evidence changes with each task

2. Can the model recognize when it needs information from another agent? How can it tell when to ask for help?